

Context

session-db is a single primary with one replica and no horizontal split.

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Goals

Remove the write hotspot and bound token validation latency.

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Alternatives

Vertical scaling, caching only, or sharding. Caching alone was rejected.

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Target topology

The target topology has exactly 3 database nodes: one shard primary per tenant band, each fronted by the authentication service and the ledger reader.

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Migration

Dual-write, backfill, cut over one tenant band at a time.

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